

1. A body moves in a straight line such that $x = 4t^2 - t^3$. Find
 - a. x when $t = 3$
 - b. v when $t = 2$
 - c. a when $t = 1$
2. Initially (i.e. when $t = 0$), the velocity of a body is 3 ms^{-1} .
 - a. If the body were accelerating at a constant rate of 4 ms^{-2} , what would its velocity be after 2 s?
 - b. Actually, the body moves in a straight line but with *variable* acceleration such that $a = 4t$. Assuming again that at $t = 0$, $v = 3 \text{ ms}^{-1}$,
 - i. find a when $t = 5$
 - ii. find v when $t = 2$
3. A body moves in a straight line such that $v = 2t^2 - 11t + 14$. Initially the displacement of the body from some fixed point O on the line is 50 m. Find:
 - a. the initial velocity of the body
 - b. the values of t when the body is at rest
 - c. the acceleration of the body when $t = 5$
 - d. the displacement of the body from O when $t = 6$.
4. A particle moves along the x -axis with position given by $x = t^3 - 9t^2 + 24t$.
 - a. Find the times, if any, when the particle is at rest.
 - b. Find the displacement of the particle over the interval $1 \leq t \leq 6$.
 - c. Find the distance travelled by the particle over the interval $1 \leq t \leq 6$.
5. A particle moves along the x -axis with position given by $x = t^3 - 2t^2 + 2t$.
 - a. Find the times, if any, when the particle is at rest.
 - b. Find the displacement of the particle over the interval $1 \leq t \leq 6$.
 - c. Find the distance travelled by the particle over the interval $1 \leq t \leq 6$.
6. A particle moves in a straight line with an acceleration $a \text{ ms}^{-2}$ at time t seconds where $a = -\frac{1}{t^2}$.
If when $t = 1$ the particle has a velocity of 3 ms^{-1} find the velocity when $t = 4$ and show that the velocity approaches 2 ms^{-1} as t increases.
7. A particle moves along the x -axis with acceleration given by $a = 14 - 6t$.
At $t = 0$, the particle is at position $x = 9$ and is moving in the *negative* x direction with speed 15 ms^{-1} .
 - a. Find an expression for x in terms of t .
 - b. Find the times, if any, when the particle is at O.
 - c. Give a careful sketch of a graph of x against t for $0 \leq t \leq 5$. [You need not find the coordinates of any stationary points.]
8. A particle, initially at O, moves along the x -axis with velocity $v = 7t^2 - 16t + 4$.
 - a. Find an expression for x in terms of t .
 - b. Find its position at the time(s) when the particle is instantaneously at rest.
 - c. Find any time(s) at which the particle returns to O.
 - d. Sketch a position-time graph for the motion, marking the coordinates of any axis intercepts and stationary points.
9. A particle moves along the x -axis with acceleration $a = 12t - 14$. Its initial position is $x = 50$, and its initial velocity is $v = -33$. Find its velocity at the time(s) at which it passes through the point $x = 32$.